



Screenshot of TutorVista's white-board tool

writing things down and drawing symbols or diagrams, and a regular text tool falls short here. The white-board developed at TutorVista has tools for drawing symbols for equations, lines, graphs, etc., which permit the tutor to explain in detail, with figures and diagrams.

Open source in digital education

Predominantly, TutorVista has always been a LAMP shop. The data centre and back-up data centre are in the US, where all the hosting is done. There are servers for content, database and database slaves, load balancing, and the *ejabberd* server. At present, there are about 50 servers—all Linux-based, while all the Web servers are Apache-based. "Our programs were mainly developed using PHP, but of late, we have been trying to use Java and Tomcat to build some additional tools, especially one of the new technologies coming up in Java called Wicket. This is under the open source Web framework. It is component-based and takes care of all the plumbing work that you need to do when you are developing a website. Wicket helps you keep your focus on your business problems. We are checking if we can use it to become more productive," Ramasubramani states.

Choosing the right technology

At TutorVista, the use of open source is not so much about a single platform as it is about the environment. Ramasubramani believes that the key in using open source is to know when and where to use which tool. He shares, "We have to deal with areas like sales, marketing, tutoring, operations and customer care, and all these have been built around open source. There may be some areas where PHP is right, and other areas where Java fits the bill. Now, we are in the process of evaluating areas where Java may be the right tool rather than PHP, which has been used extensively. Depending on the area of work, we have to use the right technology for the right job." If the wrong technology is used, a firm can end up incurring additional expenses—in terms of developer bandwidth, and the effort that goes into the technology. Ramasubramani explains with an example, "If you want to write a client program that does some

queries and executes some logic, it does not have to be driven off PHP. It may be great for running programs from within an Apache container, but using the PHP command-line does have its challenges. In such cases, I would rather use a language like Java. PHP's benefit is that it is not strongly typed, while Java's strength lies in being strongly typed. If you are working on prototypes and rapidly changing applications, PHP may help you. In other cases, when you want to make sure that everything is strongly typed, Java is a better option. There is no exact science behind this, but it is more about figuring out areas in which the right technology can be used."

Open source in mission-critical operations

TutorVista handles a little over one TB of data, which includes content developed in-house, sales processes, reports about visitors who got converted into customers, schedules, tutor and student details, customer care records and transaction data. Some of the tutoring sessions that students have replayed at any point of time have also been stored in the database. The main concern of TutorVista is to make sure the website is accessible 24/7. "We have to ensure that there is zero downtime when we deploy new releases. We have no offline component. The day our system goes down, it is immediately a big deal, because customers cannot access the site or get tutoring services. They will lose confidence in us and this will cost us. I do not recollect any time in the recent past when the platform was unavailable. I am sure there have been times in the past when we have had downtime, but I will not attribute it to the use of open source. I have worked on closed source software as well, and if something goes wrong with the platform, I know it is because the developer or deployer has made a mistake."

Moreover, the two white-boards that appear at both ends—the tutor's and the student's—communicate with each other via *ejabberd*. "It is an open source server that essentially implements the Jabber protocol and is often praised for its scalability and clustering features. It is the same protocol used by GTalk. Currently, the white-board uses a Flash-based component. Although we do not call it open source, once you observe the way Flash has evolved in the past few years, you will see that the significant number of people who use it, effectively provide support for it. However, we are trying to figure out how to convert this into an HTML 5 component. The only areas where we are not using open source is the white-board, which has used a Flash-based component, and audio, for which we use another Adobe product," shares Ramasubramani.

Why open source

Going the open source way has been an inevitable decision at TutorVista. For start-ups, one of the most important factors is keeping costs down. Web